

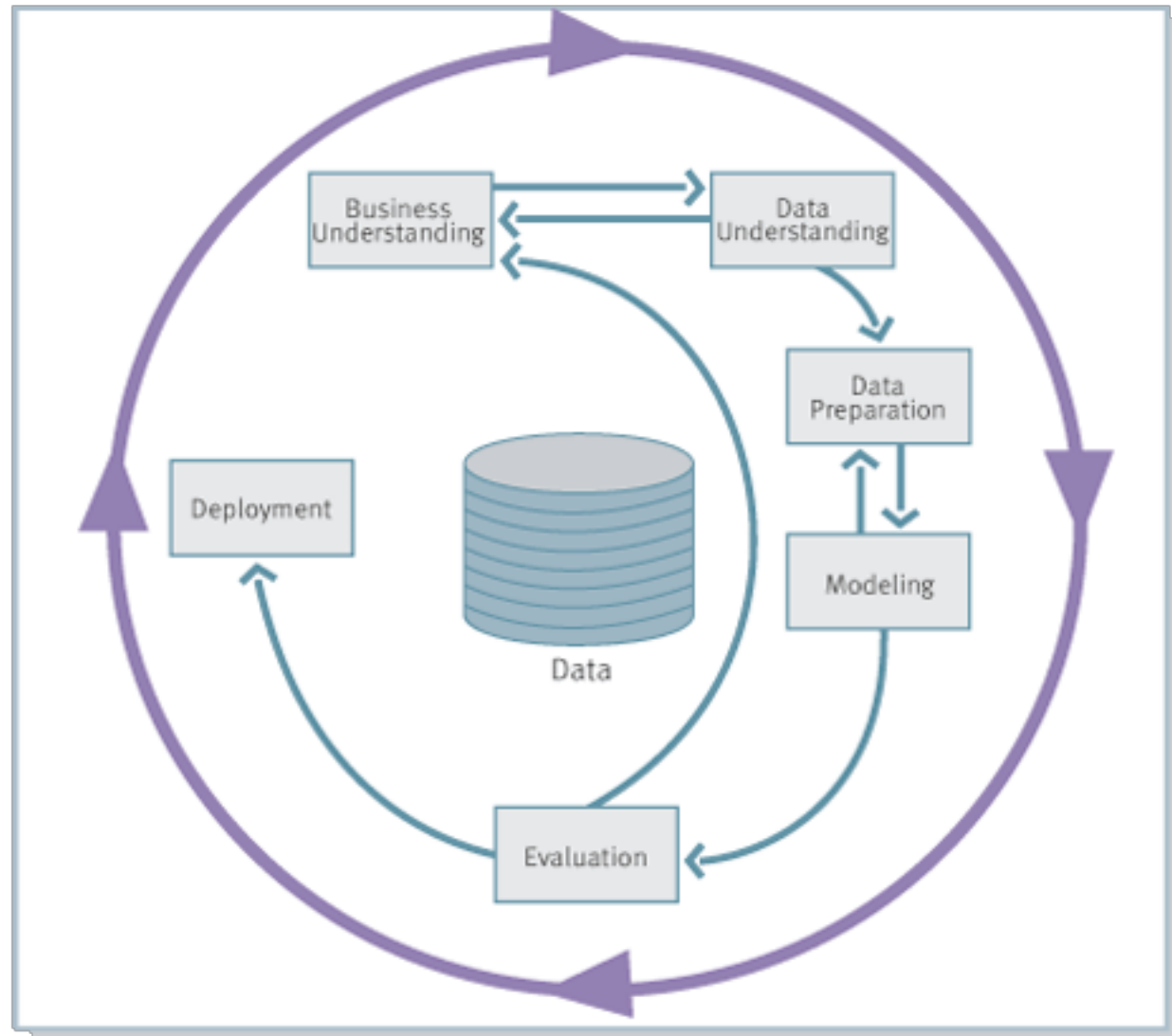
CRISP-DM

Data Mining
Howles

The Process

- Initially no standard framework to define data mining efforts
- As field matured, the need for a process and tools became evident
- Most common framework is the CRISP-DM KDD process

CRISP



The CRISP-DM

- Defines a lifecycle for data mining projects
- First step is to identify staff
 - Business experts – domain experts
 - Data administrators & data experts
 - Data miners

Phase I

Business Understanding

- As its name implies – what is the business and what are the goals?
- How will you measure success? Improved customer satisfaction, increased number of customers, cost reduction?
- Is the goal attainable?
- Can this problem be solved using data mining?
Is it doable?

Business Understanding

- What are the risks? Develop contingency plans
- What is the timeframe?
- Are appropriate resources available? People, tools, data, funding
- Can the business goals be translated into data mining goals? DM is not a silver bullet solution for all problems
- Develop a project plan

Examples

- A grocery store chain wants to send targeted coupon mailings
 - How would you do this? What would you need to know?
- Amazon recommends purchases – “customers who bought X also bought Y”
 - What does Amazon know in order to offer this service?

Addressing Business Objectives

- The health club
 - Wants to administer a survey to determine customer satisfaction
 - Intends to use the survey results to target membership renewal offers
 - Who should receive the membership incentive offer?

Addressing Business Objectives

- Your cell phone provider
 - Who is targeted to receive incentives to renew?
 - Buy new phones

Addressing Business Objectives

- The IRS
- Has several options for dealing with delinquent tax payers
 - Soft letter
 - Hard letter
 - Phone Call
 - Collection visit
- Which to use?

Phase II

Data Understanding

- Goals
 - Become familiar with the data
 - Identify data problems or inconsistencies
 - Gain some initial insights into data relationships
 - “Explore” the data

Phase II

Data Understanding

- Is everyone using the same terminology?
- What is the data source?
 - Internal?
 - Purchased?
 - Data warehouse?
- How much data is available?
- How old is the data?

Phase II

Data Understanding

- Data Quality
- What should be done with missing data values?
- Is there a way to identify incorrect values?
Ranges, valid responses?

Exploring the Data

- Determine “interesting” attributes
 - There could be hundreds of attributes – how could we do this?
- Perform some basic analysis of the data
 - Will the data provide what’s needed – is it usable?
 - Is the data of sufficient quality and quantity?

Basic Analysis

- What exactly are we looking for?
 - Excessive skew?
 - Normal distribution?
 - Descriptive statistics?
 - Data values that don't seem to make sense

Basic Analysis

- Are there attributes we can ignore? Which are most important?
- How much data is missing? What are our options?
- We'll explore these questions further in coming classes

Phase III

Data Preparation

- Often collectively called “Data Cleaning” or “Data Scrubbing”
 - Consists of cleaning, constructing (to create new features), integrating (if multiple data sources), reformatting (when > 1 format in the raw data)
 - We’ ll spend time looking into these activities

Data Preparation

- One of the most difficult, time consuming and costly phases
- If not done correctly, can compromise entire effort

Managing the Data

- Policies must exist to manage the data
- What is needed?

Phase IV

Modeling

- Many models and tools available
- Which fits best?
 - We'll answer this question as we work through the quarter

Phase IV

Modeling

- General issues with model selection
 - Which fits best with my business goals?
 - Which works best with the type and volume of data available?
 - Which will be easiest to explain to management and support in the future?
 - Which will be easiest to assess?

Phase IV

Modeling

- Model Terminology
 - Supervised Learning
 - Independent variables or predictors → Dependent classification
 - An answer is known, based on previous trials or cases
 - Is this transaction fraudulent? Is image a person or animal?
 - Unsupervised Learning
 - Result is not a previously known classification or output
 - Does this data cluster? What types of products will sell well with high-end clients?

Phase IV

Modeling

- Arguably, 3 more common tools:
 - Decision trees
 - Easy to relay results because output is rule-based
 - Linear or logistic regression
 - Neural nets
 - Difficult to build and train models; difficult to relay to management but tend to be accurate
- But – try them all (and others). There is no “best” for all situations

Phase V

Evaluation ->> Deployment

- How accurate are the results?
- Did we meet the initial business goals?
- Is it cost effective?
- Are there other business goals or projects where this model could apply?
- This is an iterative process. May need to build several models to find the best fit

Continuous Improvement

- Post mortem – failures? Extra work? Better approaches?
- What are the next steps?

Let's Discuss

- How does the CRISP-DM model differ from the SLC?
- Requirements make or break a software project. Is there an equivalent Achilles' Heel for data mining?
- Does this model need some type of configuration management? How might CM be helpful? (see next)

CM?

- Not specifically mentioned in CRISP-DM, but seems like Some sort of archive and/or historical documentation. In SW Development, we have CM. What do we have for DM?
 - Issue: In 2003, your DM model identified Mr. Jones as a safe mortgage risk. He just defaulted. What went wrong – your model? Something with Mr. Jones? Did you have bad data that was used to make the decision?
 - Without some sort of CM, you don't have an easy way to look back

Other Elements

- Ethics?
 - Safety of the data
 - Treatment of sensitive data
- Others

Resources

- kdnuggets.com
- kdd.org